जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

04th March 2025

Roll No. | | | | | | | | | | | | |

SEMESTER MIDTERM EXAMINATIONS

Programme	:	**B. Tech. CSE & CSE Cyber**	**Semester**	:	**VI**
Course Name	:	**Deep Learning**	**Course Code**	:	**BECSE3C008, BECCS3C008**
Duration	:	**1.5 Hours**	**Credits**	:	**03**
Maximum Marks		**45**	**Minimum Marks**	:	-

After pursuing this course, the student will be able to:	
CO 01	To understand the fundamental concepts of deep learning and its architecture.
CO 02	Design, train and optimize neural networks using advanced techniques.
CO 03	Use regularization and optimization techniques to improve model performance.
CO 04	Implement various deep neural networks such as Convolutional and Recurrent Neural Networks, Transformers etc.
CO 05	Apply deep learning to solve real-world problems and evaluate model results effectively.

Part A **Very Short Answer Type Logical Questions of 04 marks related to different COs**		**Total Marks 5x2=10**	
S. No.	**Statement**	**Levels**	**CO**
A1.	Differentiate between the McCulloch-Pitts Neuron and Perceptron?	L2	CO-01
A2.	What is the role of the learning rate in the Gradient Descent algorithm?	L2	CO-02 CO-03
A3.	Write the formula for Cross-Entropy loss with proper comments?	L2	CO-03
A4.	Compute the Mean Squared Error (MSE) between actual values (y) and predicted values ($\hat{y}$) for the sigmoid neuron with input vector X= [4,3,2,1], output vector O= [0.8] and weight vector [0.5, 0.6, 0.2, 0.1].	L3	CO-01 CO-02
A5.	How does Deep Learning differ from Machine Learning?	L1	CO-02

PART B **Short Answer Type Analytical Questions of 6 marks related to different COs** **(Attempt any Five)**		**Total Marks 5x3=15**	
B1.	Solve the XOR problem using a Multi-Layer Perceptron (MLP) and brute force optimization technique.	L4	CO-01
B2.	Design a Feedforward Neural Network (FFNN) with 4 layers, where the input layer contains 10 neurons, each hidden layer contains 7 neurons and the output layer has 4 neurons. Computer the number of parameters (weights and bias) in the complete network and in between individual layers?	L3	CO-02
B3.	Discuss the significance of sigmoid neuron and how it differs from perception.	L4	CO-02

B4.	Given a dataset of 10,000 samples, batch sizes of 1000, 2000, 3000, and 5000. Compute the number of iterations required for 60 epochs using: a)Vanilla Gradient Descent b)Stochastic Gradient Descent (SGD) c)Mini-batch Gradient Descent	L5	CO-01
B5.	What are the various goals of optimization in Deep Learning?	L3	CO-01
B6.	Implement AND and OR logic functions using McCulloch-Pitts neuron providing appropriate threshold values.	L2	CO-03
	Part C **Long Answer Type Descriptive Questions of 20 marks related to different COs** **(Only Two questions to be attempted)**	**Total Marks 2x10=20**	
C1.	a) Write and explain the Perceptron Learning Algorithm for updating the parameters in MLP. b) Consider a simple neural network with one hidden layer with 3 neurons where you can use Sigmoid function as the activation function and Mean Square Error (MSE) as the loss function and input layers comprises of the two nodes, X1 and X2 where X1 = 2.8 and X2 = 1.7 and target output, y = 0.7. Initialize weights and bias between -1 and 1. Perform ~~two~~ one steps of back-propagation and update the weights and bias values accordingly?	L4	CO-01
C2.	a) Describe the Backpropagation Algorithm with Gradient Descent for training feed forward neural network? b) Derive how gradient can be computed as a function of predicted value, actual value and corresponding input in single neuron.	L3	CO-02
C3.	Discuss the mathematical equations and intuition behind following optimization algorithms? a) Stochastic Gradient Descent (SGD) b) Momentum based Gradient Descent c) Nesterov Accelerated Gradient Descent (NAG) d) AdaGrad Compare their strengths and weaknesses, and explain their impact on deep learning model performance?	L4	CO-01 CO-02 CO-03